Prabhjeet Singh

New York, NY (Open to relocation) | +1 (718) 551-4498 | prabhjeet2570@gmail.com | LinkedIn | GitHub

Education

New York University — New York, NY
Master of Science in Computer Science | GPA: 4.00/4.00 — *Sept. 2025 – May 2027 (expected)*
- **Coursework:** Artificial Intelligence | Machine Learning | Big Data | Software Engineering | Deep Learning | Language Models, Reinforcement Learning, Reasoning | Design and Analysis of Algorithms

Netaji Subhas Institute of Technology — New Delhi, India
Bachelor of Engineering in Information Technology — *Aug. 2018 – May 2022*

Experience

Software Engineer — July 2022 – August 2025
DP World — *Gurugram, India*
- Implemented a configurable post-invoice workflow system for **22 clients**, coordinating 12 downstream integrations with scheduling, dependency controls, configurable retries, and email alerts after retry exhaustion
- Reduced proforma-to-invoice conversion p95 response time from approximately **7 seconds to 800 ms** by eliminating redundant billing-data duplication
- Engineered a Kafka-based asynchronous proforma invoice pipeline from client requirements and product input, owning the complete technical design and implementation to replace synchronous processing
- Integrated 7 shipping-line APIs with the in-house cargo-offer system to broaden offer availability, and streamed offers through Server-Sent Events as each source responded instead of waiting for all providers
- Developed a credit note system that replaced manual refund requests with financial-system integration for automated wallet reversals, with **1,500 credit notes** generated in the first year after production rollout

Graduate Student Researcher — September 2026 – Present
New York University, BAAHL Lab — *New York, NY*
- Evaluate MoViNet-A0 on Kinetics-600 in PyTorch, measuring classification accuracy after individual and cumulative activation removals to investigate reducing costly nonlinear operations in encrypted inference

Deep Learning Course Assistant — September 2026 – Present
New York University — *New York, NY*
- Support a 150-student Deep Learning course through office hours, assignment grading, and written technical feedback

Projects

Huddle | *Python, FastAPI, LangGraph, PostgreSQL, Docker* — GitHub
- Designed a LangGraph study-room booking assistant with reservation approval, workflow state saved in PostgreSQL, booking conflict prevention, and an outbox worker for retries and simulated calendar reconciliation

FinSight | *Python, FastAPI, React, TypeScript, SQLite* — GitHub
- Built a full-stack SEC filing research application with Ollama-powered retrieval-augmented generation (RAG), hybrid BM25/MiniLM retrieval, cross-encoder reranking, source citations, and deterministic financial calculations

AeroStream Analytics | *Python, Flask, Kafka, PySpark, Parquet, Docker* — GitHub
- Created an aviation dashboard with Kafka/OpenSky live traffic maps and single-node PySpark/Parquet analysis of 9.57M historical flight records for route, airport, and estimated CO_2 summaries

SmolVLM ScienceQA | *Python, PyTorch, Transformers, PEFT, DoRA* — GitHub
- Fine-tuned SmolVLM-500M-Instruct for visual science questions with DoRA and 4.79M trainable parameters, balanced answer-choice augmentation, validation-selected checkpoint ensembling, and selective permutation averaging

Technical Skills

Languages: Python, Java, SQL, JavaScript/TypeScript, C++
Web & APIs: Spring Boot, FastAPI, Django, Flask, React, REST APIs, GraphQL, Server-Sent Events (SSE)
Databases & Data Engineering: PostgreSQL, MongoDB, SQLite, Redis, Apache Kafka, PySpark, Parquet
ML & AI: PyTorch, Hugging Face Transformers, PEFT (LoRA, DoRA), LangGraph, RAG, NumPy
Cloud & DevOps: AWS Elastic Beanstalk, Azure Kubernetes Service (AKS), Docker, Azure DevOps, GitHub Actions
Developer Tools & Testing: Git, Linux, Vite, JUnit, pytest, Playwright, Claude Code, GitHub Copilot, OpenAI Codex